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Genome-wide identification, characterization and expression analysis of the chalcone synthase gene family in Chinese cabbage

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Abstract

Background Chalcone synthase (CHS) is a key rate-limiting enzyme in the flavonoid synthesis pathway. Flavonoids are crucial secondary metabolites that play significant roles in plant growth, development, and stress resistance. The *CHS* gene (*BrCHS*) family in Chinese cabbage has not yet been studied.

Results We identified 10 *BrCHS* genes distributed across 7 chromosomes in the Chinese cabbage genome. Their encoded proteins all contain the Chal_Sti_Synt_C (PF02797) and Chal_Sti_Synt_N (PF00195) domains and can be classified into two groups based on systematic evolution analysis. These *BrCHS* genes contain 2–4 exons and numerous *cis*-acting elements responsive to light, hormones, stress, growth and development in the *BrCHS* gene promoters. We also revealed that the expression of *BrCHS2* and *BrCHS8* increased under treatment with methyl jasmonate, salt, or drought stress. Virus-induced gene silencing (VIGS) of *BrCHS4* inhibited the expression of *BrCHS4* and reduced the flavonoid and anthocyanin contents in leaves.

Conclusions Ten *BrCHS* family genes are present in the genome of Chinese cabbage. These *BrCHS* genes seemingly maintained similar characteristics and functionalities during evolution. Our results demonstrated that *BrCHS4* is involved in flavonoid and anthocyanin accumulation in Chinese cabbage and identified candidate genes for purple Chinese cabbage breeding.

Keywords Chinese cabbage, Chalcone synthase gene family, Genome-wide identification, Characterization and expression analysis

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Background

Chinese cabbage (*Brassica rapa* L. ssp. *Pekinensis*), a species in the genus *Brassica* of the family Cruciferae, is an important vegetable crop in China. Chinese cabbage leaves are the main product and are primarily green, yellow, orange or purple. The color of leaves is also a main agronomic trait and can affect leaf photosynthetic efficiency and yield. Chlorophyll, carotenoids, and anthocyanins are widely present in the leaves, flowers, stems, roots and fruits of plants, and the proportion and content



of these three compounds determine the color of these plant tissues and organs [1, 2]. The contents of chlorophyll, carotenoids, and anthocyanins in different Chinese cabbage varieties vary, leading to different leaf colors, and providing different nutritional values. Indeed, Chinese cabbage leaf color development is closely related to the accumulation of anthocyanins. Anthocyanins are water-soluble pigments that exist mainly in the form of glycosides and can form various colors such as red and blue. They play important roles in plant growth, development, and defense [3–5]. The accumulating anthocyanins in vacuoles leads to different colors in different tissues (including flowers, leaves and pericarps) [6]. Anthocyanins are involved in plant resistance to biotic and abiotic stresses, including pathogen infection, ultraviolet radiation and low temperature [7, 8]. At present, the main color of Chinese cabbage on the market is green, whereas other colors of Chinese cabbage have not yet become popular. There is an urgent need to cultivate varieties of Chinese cabbage that are rich in anthocyanins.

Chalcone synthase (CHS) is the key enzyme of the anthocyanin pathway [9]. Anthocyanins are flavonoid compounds derived from phenylalanine [10]. Flavonoid compounds are complex and diverse secondary metabolites produced in plants. They play important roles in flower color formation and pathogen infection prevention in plants [11]. Many key genes are involved in flavonoid synthesis in plants. The first key enzyme in the flavonoid biosynthesis pathway is CHS. The substrate catalyzed by CHS in most plants is coumarin coenzyme A, while in a few plants, CHS can utilize cinnamoyl coenzyme A (cinnamoyl CoA) and the coffee phthalein coenzyme A (caffeoyl CoA) as reaction substrates. It can catalyze the combination of 3 molecules of malonyl-CoA and 1 molecule of p-coumaryl-coenzyme A to produce 1 chalcone. This product is also the basis for the synthesis of other flavonoids [12, 13].

The CHS family contains many CHS and CHS-like proteins [14]. CHS mainly catalyzes the condensation of cinnamoyl CoA and malonyl-CoA to form chalcone, which is different from CHS-like in substrate specificity [15]. With respect to amino-acid phylogeny, CHS and CHS-like proteins formed clades according to enzyme function. There are some similarities in the amino acid sequences of the two, but there are also differences. These differences may lead to differences in their spatial structure and function [16]. Studies have shown that CHS-like proteins (Less adhesive protein LAP5 and LAP6) in *Arabidopsis thaliana* affect both the synthesis of pollen fatty acids and exine development [17]. The most common CHS and CHS-like proteins in plants are polyketide synthase (PKS) type III proteins [16]. PKS can catalyze the formation of polyketide compounds into stilbenes,

flavonoids, antibiotics, mycotoxins and other substances and plays a vital role in preventing UV radiation, plant vegetative growth and the response to pathogen infection [18, 19].

The *CHS* genes constitute a polygenic family with strong conservation and encode a protein of approximately 400 amino acids. *CHS* genes have been cloned or identified from many plants. There are 4 *CHS* gene members in the model plant *Arabidopsis* [20]. Fourteen *CHS*s have been identified in maize (*Zea mays*) [21] and soybean (*Glycine max*) [22]. In addition, the tobacco varieties Honghua Dajinyuan [23], purple petunia [24], petunia [25], gerbera [26] and pea [27] contained 5, 6, 8, 3 and 8 *CHS* genes, respectively; moreover, a total of 27 *CHS* genes were found in rice [28]. Although *CHS* has been studied in depth at whole-genome the level in multiple species, research on Cruciferous crops, especially in the important commercial crop Chinese cabbage. *CHS* genes expression are affected by developmental and environmental factors such as light, adverse conditions, transcription factors and fruit maturation stage [29]. CHS is not only related to anthocyanin synthesis but also an endogenous negative regulator of auxin transport [30], and regulates the development of plant organs, such as floral organs [31]. In addition, the expression of *CHS* gene family members is tissue specific [32], and the expression levels at different growth stages also differ [22].

The sequences of Chinese cabbage genome have been published [33], which allows us to identify and study the *CHS* gene family of Chinese cabbage at the whole-genome level. In this study, the *BrCHS* family of Chinese cabbage was identified and analyzed via bioinformatics technology, and how the *BrCHS* genes respond to hormones and stress in Chinese cabbage was also explored using fluorescence quantitative PCR. Our results provide a foundation for elucidating the functions of the *BrCHS* family in Chinese cabbage.

Materials and methods

Plant materials and treatment

Chinese cabbage (*Brassica rapa* L. ssp. *Pekinensis*, a high-quality hybrid variety of purple Chinese cabbage, 'ZiYi', Kwonong Seed Co.) was used as the test material. Purple Chinese cabbage seeds were chosen and sown in a nutrient bowl (9 cm × 9 cm). Routine seedling management was conducted in the climate chamber at a temperature of 25 ± 2°C, humidity of 50%–60%, light intensity of 220 μmol·m⁻²·s⁻¹, light duration of 16 h, and darkness of 8 h.

Identification of *BrCHS* genes in Chinese cabbage

All of the complete gene and protein sequences of *B. rapa* L. ssp. *pekinensis* (Chinese cabbage) and *Brassica*

oleracea var. *capitata* were downloaded from the Brassicaceae Database (BRAD, <http://brassicadb.cn>) [34]. We also downloaded Arabidopsis genome data from the Arabidopsis Information Resource (TAIR, <https://www.arabidopsis.org/index.jsp>). Hidden Markov model (HMM) profiles (PF00195 and PF02797) were obtained from the Pfam protein family database (<http://pfam.xfam.org/>) [35], and the Chinese cabbage BrCHS protein family members were identified using the simple HMM search function in TBtools software [36], with a maximum E-value of 1×10^{-5} . Pfam was subsequently used to authenticate each BrCHS protein. In addition, we predicted the physical and chemical parameters of each protein using the Swiss Institute of Bioinformatics (SIB) (ExPASy, <https://web.expasy.org/protparam/>) tool.

Analysis of conserved motifs and gene structure

The MEME program (version 5.4.1, <https://meme-suite.org/meme/tools/meme>) [37] was used to identify the conserved motifs of the BrCHS amino acid sequences with the following parameters: zero or one occurrence per sequence (zoops), a maximum number of motifs of 10 and a motif width of 6–50. The conserved domains of the BrCHS proteins were determined via Batch CD-Search (NCBI: <https://www.ncbi.nlm.nih.gov/guide/domains-structures/>). The gene structure of the *BrCHS* genes was visualized using the Gene Structure View (Advanced) package in TBtools.

Prediction of the BrCHS protein structure

SOPMA (<http://npsa-pbil.ibcp.fr/>) and SWISS-MODLE (<https://swissmodel.expasy.org/>) [38] were used to predict the secondary and tertiary structures of proteins, respectively. WoLF PSORT Prediction (<https://wolfpsort.hgc.jp/>) was used to predict the subcellular localization information of the CHS proteins. HMMTOP (<http://www.enzim.hu/hmmtop/index.php/>) was used to predict transmembrane helices and the topology of proteins. NetPhos 3.1 server online software (<https://services.healthtech.dtu.dk/service.php?NetPhos-3.1>) was used to predict and analyze protein phosphorylation sites.

Chromosomal locations and phylogenetic and evolutionary analysis

A physical map of the *BrCHS* genes was drawn by using TBtools according to their chromosomal positions in Chinese cabbage. The CHS protein sequences of Chinese cabbage, *Brassica oleracea*, Arabidopsis and *Solanum melongena* [25, 39] were aligned using the MEGA version 11.0 program [40]. A phylogenetic tree was constructed to analyze the evolutionary relationships of the CHS proteins using the maximum likelihood method with 1000 bootstrap replicates by the IQ-TREE web

server (<http://iqtree.cibiv.univie.ac.at/>) and the best-fit model according to the Bayesian information criterion (BIC): mtREV24. The phylogeny based on the BrCHS proteins was visualized using the Interactive Tree of Life (iTOL: <https://itol.embl.de/>). The multiple collinearity scan toolkit (MCScanX) in TBtools was used for collinear analysis. The collinear relationships of the *BrCHS* genes were visualized using the advanced Circos tool in TBtools. The Search Tool for the Retrieval of Interacting Genes/Proteins (STRING) (<http://string-db.org/cgi>) database was used to construct an interaction network among the BrCHS proteins.

Analysis of cis-acting regulatory elements (CAREs) in BrCHS

The upstream sequences (2 kb) of the *BrCHS* coding sequences in Chinese cabbage were retrieved from the genome sequence and subsequently submitted to PlantCARE (Database of Plant *Cis*-acting Regulatory DNA Elements, <http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) to identify regulatory elements [41].

RT-qPCR of BrCHS genes under phytohormone and abiotic stresses

When the plants had four leaves, they were irrigated with water (CK, control), 0.2 mmol·L⁻¹ methyl jasmonate (MeJA), 400 mmol·L⁻¹ NaCl (salt treatment) and 20% PEG6000 (drought treatment). Samples were taken 12 h after each treatment for three replicates. Chinese cabbage *β-actin* was chosen as an internal reference gene, and Primer 5.0 was used to design specific primers (Supplemental Table 1). Three technical replicates were performed for each reaction. The difference in target gene expression was analyzed using the 2^{-ΔΔCT} method [39].

Instantaneous silencing of BrCHS4 gene in Chinese cabbage

For the virus-induced gene silencing (VIGS) assay, the specific 40 bp *BrCHS4* hairpin fragment was inserted into Eco105I (SnaBI) of the pTY-S VIGS vector to generate the pTY-*BrCHS4* vector. Using pTY-BrPDS as a positive control, pTY-S, pTY-*BrCHS4* and pTY-BrPDS were transferred into Chinese cabbage seedlings by bombarding them with gene gun. The seedlings were sampled for phenotypic observation, flavonoid, anthocyanin and phenol determination, and gene expression analysis 20 days after bombardment. At least 10 Chinese cabbages were bombarded for each strain with the corresponding vector, and the instantaneous silencing assays were repeated three times.

Measurements of flavonoid, anthocyanin and total phenol contents

Approximately 200 mg of Chinese cabbage leaf were frozen in liquid nitrogen and ground into powder, and then 50 mL methanol solution containing 1% (v/v) HCl was added in the tube and incubated at 4°C in the dark until it faded. With 1% HCL-methanol solution as a blank control, the absorbance of the supernatant was measured at 280, 325, 600 and 530 nm, and the process was repeated three times.

Statistical analysis

In the RT-qPCR and pigment content determination, statistical analysis (ANOVA) and statistical significance ($P<0.05$) were performed using the SPSS program, version 26.0. Following the methodology of a prior study by Ahmad et al. [42], the fluorescence quantification data and pigment content were analyzed and expressed as means of the standard deviation (SD) of the three biologically independent replicates for all measured parameters. GraphPad Prism (version 8.0.1) was used for graphical representation.

Results

Identification and physicochemical property analysis of chalcone synthase genes in Chinese cabbage

By searching the Pfam database, we confirmed all the candidate proteins encoded by *CHS* gene had the Chal_sti_synt_C (PF02797) and Chal_sti_synt_N (PF00195) domains. A total of 10 *BrCHS* genes were identified in Chinese cabbage, named *BrCHS1-BrCHS10* according to their chromosomal location (Fig. 1). These ten *BrCHS*

genes were unevenly distributed on seven chromosomes. The number of *BrCHS* genes on chromosome A03 was the greatest with 3. *BrCHS2* and *BrCHS3* were distributed on chromosome A02. There was only one gene on chromosomes A01, A04, A08, A09, and A10, and there was no *BrCHS* gene on chromosomes A05, A06 and A07. This indicates that the *CHS* gene in Chinese cabbage may play an important role in the entire genome. In addition, 11 *CHS* genes and 4 *CHS* genes from cabbage (*B. oleracea* var. *capitata*) and Arabidopsis were characterized.

We then predicted the physicochemical properties of *BrCHS*. The number of amino acids encoded by the 10 *BrCHS* genes ranged from 392 to 396. The molecular weights of the *BrCHS* proteins was between 42801.08 and 43239.97 Da, and the theoretical isoelectric point was between 5.60 and 6.61, all of these proteins are acidic proteins. The instability indices of the 10 *CHS* proteins in Chinese cabbage ranged from 33.03 to 47.87. Among them, 2 members (*BrCHS5* and *BrCHS6*) were unstable proteins with instability indices greater than 40, and the remaining 8 were stable proteins. The aliphatic indices of the *BrCHS* proteins ranged from 88.09–96.79. The GRAVY values of *BrCHS* proteins which were all negative, indicated that these proteins were hydrophilic (Table 1).

Phylogenetic analysis of BrCHS proteins

In order to investigate the phylogenetic relationships among members of the *CHS* family, we constructed a phylogenetic tree using *CHS* proteins from Chinese cabbage, cabbage belonging to the same genus as Chinese cabbage, *Arabidopsis thaliana* belonging to the same

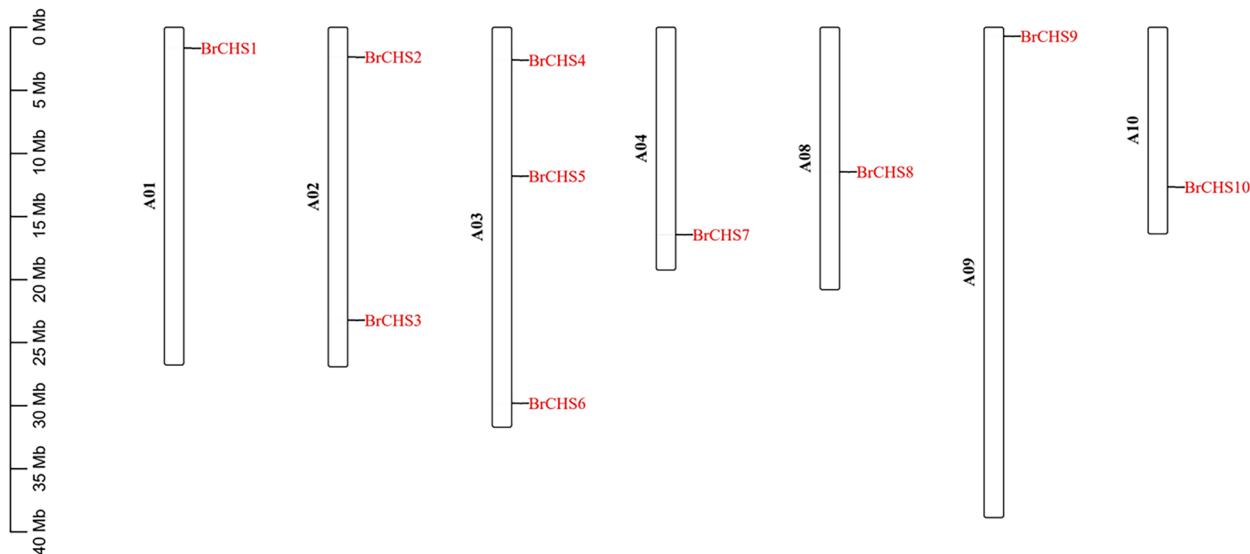


Fig. 1 Locations of the 10 *CHS* genes on Chinese cabbage chromosomes. The ruler on the left represents the length of the chromosome

Table 1 Features of the *BrCHS* genes identified in Chinese cabbage

Gene name	Gene ID	Number of amino acids(aa)	Molecular weight (Da)	Theoretical pI	Instability index	Aliphatic index	GRAVY
<i>BrCHS1</i>	Bra011566	392	42,835.13	5.61	37.23	90.56	-0.200
<i>BrCHS2</i>	Bra023441	394	42,952.67	6.61	34.73	90.58	-0.095
<i>BrCHS3</i>	Bra020688	394	43,019.87	5.86	35.87	93.53	-0.008
<i>BrCHS4</i>	Bra006224	396	43,239.97	6.57	35.46	90.38	-0.099
<i>BrCHS5</i>	Bra000559	392	42,931.23	5.82	42.75	88.09	-0.204
<i>BrCHS6</i>	Bra017681	392	42,802.03	6.15	41.72	89.06	-0.219
<i>BrCHS7</i>	Bra017147	388	42,732.22	6.08	39.95	92.22	-0.192
<i>BrCHS8</i>	Bra034658	392	42,801.08	5.60	38.77	91.81	-0.174
<i>BrCHS9</i>	Bra036307	393	43,179.01	6.47	33.03	96.79	-0.036
<i>BrCHS10</i>	Bra008792	395	42,982.48	5.92	33.23	90.13	0.096

family as Chinese cabbage, and eggplant, a plant in the *Solanaceae* family that is distantly related to Chinese cabbage (Fig. 2). The *BrCHS* proteins were classified into two major groups (CHS and CHS-like classes) based on the phylogenetic tree, with bootstrap support values. *BrCHS2*, *BrCHS3*, *BrCHS4*, *BrCHS9* and *BrCHS10* belong to the CHS class, and the CHS-like class consist

of *BrCHS1*, *BrCHS5*, *BrCHS6*, *BrCHS7* and *BrCHS8*. Dobritsa predicted that the LAP5 (AT4G34850) and LAP6 (AT1G02050) proteins are enzymes in the CHS family of *Arabidopsis thaliana* [20]. In addition to these two proteins and the well-characterized AtCHS TT4 (AT5G13930) [43], *Arabidopsis* contains a third CHS-related gene, AT4G00040. LAP5 (AT4G34850), LAP6

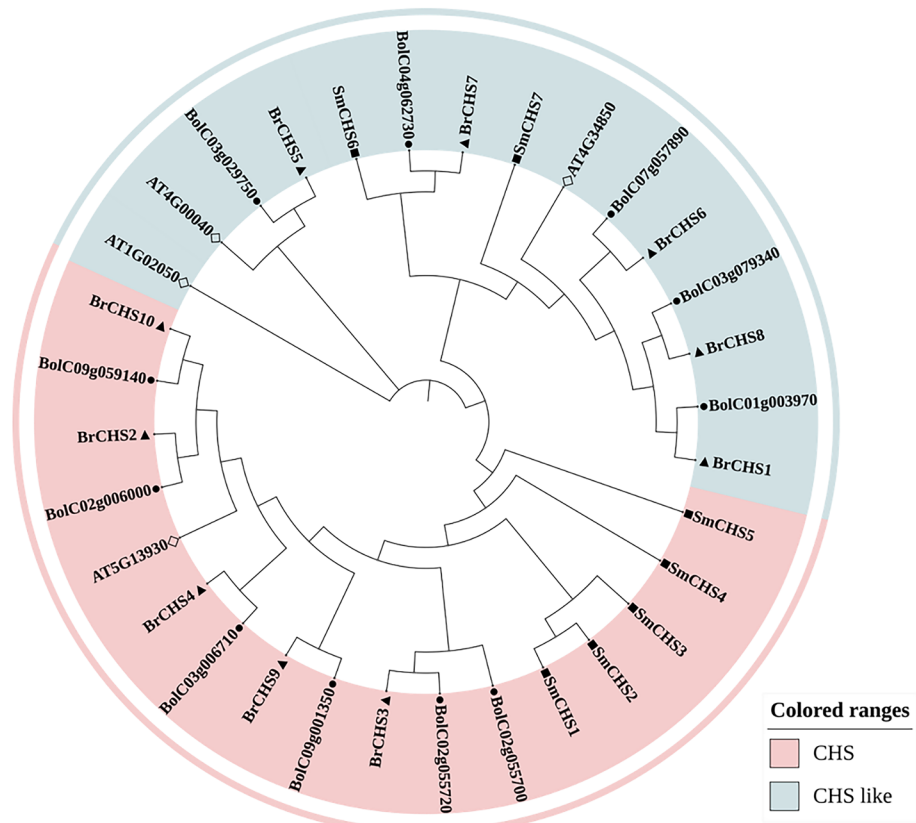


Fig. 2 Phylogenetic tree of CHS proteins in Chinese cabbage (▲), *B. oleracea* (●), *Arabidopsis* (◇) and *Solanum melongena* L (■). Pink represents CHS class proteins and light blue represents CHS-like class proteins

(AT1G02050) and AT4G00040 are classified as CHS-like in phylogeny. In addition, six cabbage genes belonged to the CHS class, and the other five belonged to the CHS-like class. Among the seven eggplant SmCHS proteins, five belong to the CHS class and two belong to the CHS-like class.

BrCHS protein motif and *BrCHS* gene structure

Gene structure analysis revealed 2 exons in the 7 *BrCHS* genes, 3 exons in the *BrCHS1* and *BrCHS3* genes, 4 exons in the *BrCHS8* (Fig. 3A). The results revealed that

BrCHS genes possess varied exon lengths and numbers. Ten conserved motifs were investigated in the CHS proteins of Chinese cabbage. The detailed information of the 10 motifs is listed in Fig. 3B and D. Among them, there were 9 conserved motifs in *BrCHS1*, *BrCHS5*, *BrCHS6*, *BrCHS7* and *BrCHS8*, and the arrangement was the same. There were 10 conserved motifs in *BrCHS2*, *BrCHS3*, *BrCHS4*, *BrCHS9* and *BrCHS10*, and the arrangements were consistent. These two groups of proteins only differ by Motif 9. This suggests that Motif 9 is a key motif that distinguishes Chinese cabbage CHS



Fig. 3 BrCHS proteins motif analysis and gene structure. **A** BrCHS proteins phylogenetic relationship and gene structure. The exons are shown in yellow. **B** Conserved motif analysis of BrCHS proteins. **D** The boxes with different colors indicate different motifs. The sequence of each motif is shown in. **C** Conserved domain analysis of the BrCHS proteins. The conserved CHS domains are shown in the indicated colors

protein from CHS-like protein. Motif 9 can form specific domains that are essential for protein folding and maintaining a stable three-dimensional structure. Analysis of the conserved domain (Fig. 3C) revealed that all CHS proteins contain only one domain. BrCHS1, BrCHS5, BrCHS6, BrCHS7 and BrCHS8 only contain one structural domain PLN03169. Members of the anther-specific CHS-like enzyme contain the PLN03169 domain. The CHS class contains five protein domain structures (PLN03168, PLN03170, PLN03171, PLN03172, and PLN03173), and the gene copy number varies between one (*Arabidopsis thaliana*) and 70 (*A. hypogaea*) [44]. BrCHS2 and BrCHS3 contain only the structural domain of the PLN03170 superfamily. BrCHS4 and BrCHS10 contain only the structural domain of the PLN03172 superfamily, and BrCHS9 contains only the structural domain of the PLN03173 superfamily.

Prediction and analysis of BrCHS protein structure

To understand the structures of BrCHS proteins, the structures of the BrCHS proteins in Chinese cabbage were analyzed (Table 2). The analysis of the secondary structure revealed that the secondary structure was mainly composed of alpha helices, extended strands, beta turns and random coils. Among the four secondary structures, the proportion of biological secondary structures of the nine BrCHS proteins was as follows: alpha helix > random coil > extended strand > beta turn. Only one BrCHS protein (BrCHS7) exhibited the following sequence: random coil > alpha helix > extended strand > beta turn in the secondary structure. The beta turns accounted for the smallest proportion of the BrCHS proteins. The transmembrane structure

prediction showed that BrCHS1, BrCHS5, BrCHS7 and BrCHS8 all have two transmembrane helices, while the other 6 BrCHS proteins have no transmembrane structure. Subcellular localization prediction revealed that BrCHS4, BrCHS5 and BrCHS7 were localized in chloroplasts, while the rest were localized in the cytoplasm. The results of predicting the phosphorylation sites of the CHS protein in Chinese cabbage showed that serine and threonine were the most abundant phosphorylation sites, while lysine was the least abundant.

The tertiary structure of the CHS family proteins was predicted through the SWISS-MODEL website (Fig. 4). BrCHS2, BrCHS3, BrCHS4, BrCHS9 and BrCHS10 have high similarity, and BrCHS1, BrCHS5, BrCHS6, BrCHS7 and BrCHS8 have relatively low similarity with other proteins. The difference in the tertiary structure of BrCHS family proteins is due to the large difference in the number and allocation position of introns/exons in their gene structures, which leads to a large difference in the number of genomic and coding amino acids. Second, in the secondary structure of the encoded protein, different alpha helices, extended strands, beta turns and random coils lead to different spatial folds.

Evolutionary analysis of BrCHS genes

Collinearity analysis can make good use of genomic resources and effectively reveal evolutionary relationships in species [45]. Gene duplication events include tandem and segmental duplications. Genes located on different chromosomal segments are defined as segmental duplications [46], and gene clusters generated by fragments separated by 5 or fewer genes are considered tandem duplications [47]. In this study, six gene

Table 2 Structural characteristics of the BrCHS proteins

Protein name	Secondary structure				Number of transmembrane helices	Phosphorylation site number/Phosphorylation amino acid number			Subcellular localization
	Alpha helix (%)	Extended strand (%)	Beta turn (%)	Random coil (%)		Serine	Threonine	Tyrosine	
BrCHS1	44.90%	14.80%	7.14%	33.16%	2	14	14	2	cytoplasm
BrCHS2	42.89%	16.24%	6.85%	34.01%	0	16	10	2	cytoplasm
BrCHS3	41.12%	16.50%	6.60%	35.79%	0	14	13	1	cytoplasm
BrCHS4	42.93%	15.91%	6.57%	34.60%	0	19	9	3	chloroplast
BrCHS5	44.90%	16.33%	8.16%	30.61%	2	19	13	3	chloroplast
BrCHS6	40.56%	16.33%	7.14%	35.97%	0	24	15	3	cytoplasm
BrCHS7	34.79%	18.81%	0.00%	46.39%	2	13	16	5	chloroplast
BrCHS8	38.52%	17.60%	6.63%	37.24%	2	16	13	3	cytoplasm
BrCHS9	40.56%	16.33%	6.89%	36.22%	0	15	8	2	cytoplasm
BrCHS10	45.06%	14.94%	6.84%	33.16%	0	19	13	1	cytoplasm

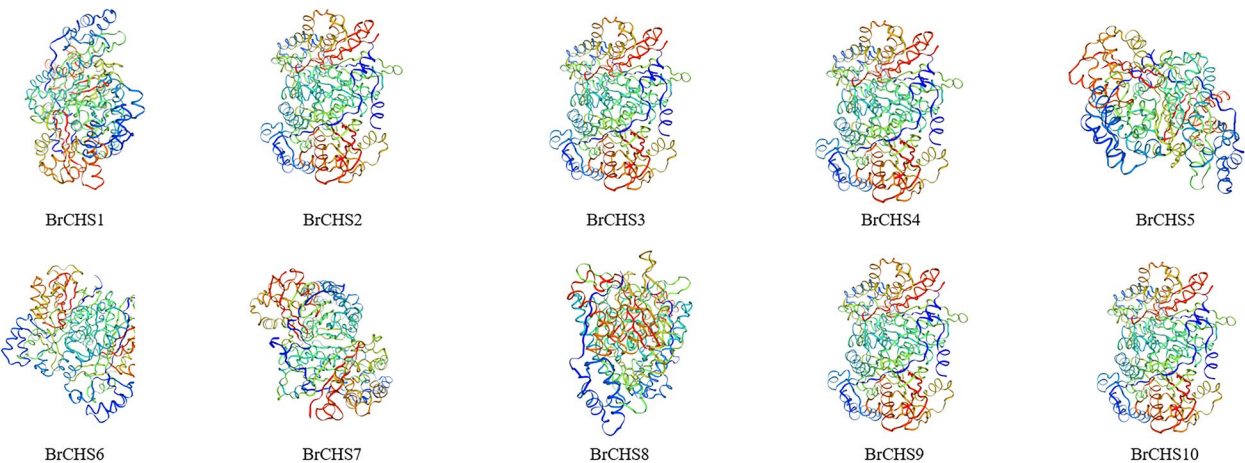


Fig. 4 Tertiary structure of the BrCHS proteins

pairs (*BrCHS1/6*, *BrCHS1/8*, *BrCHS2/4*, *BrCHS2/10*, *BrCHS6/8* and *BrCHS4/10*) were found to be involved in segmental repeats (Fig. 5A). The nonsynonymous/synonymous ratio (K_a/K_s) is usually used as a measure of selection pressure at the protein level [48, 49]. Generally, a K_a/K_s ratio >1 represents positive selection and promotes evolution. A ratio of 1 indicates neutral selection, and a ratio of <1 indicates negative selection or purifying selection. The K_a/K_s values of the six *BrCHS* gene pairs with segmental replication events were less than 1, which

indicated that the *CHS* gene family of Chinese cabbage was purified and selected and that the evolution was slow at the protein level (Table 3). These results revealed the conservation of the *CHS* gene family in Chinese cabbage.

To further examine the syntenic relationships of the *BrCHS* genes with those of the model plant *A. thaliana*, the MCScan toolkit was used to find orthologous genes between the genomes (Fig. 5B). We identified 8 orthologous gene pairs between the *BrCHS* and *AtCHS* genes. The results showed that 6 *BrCHS* genes were collinear

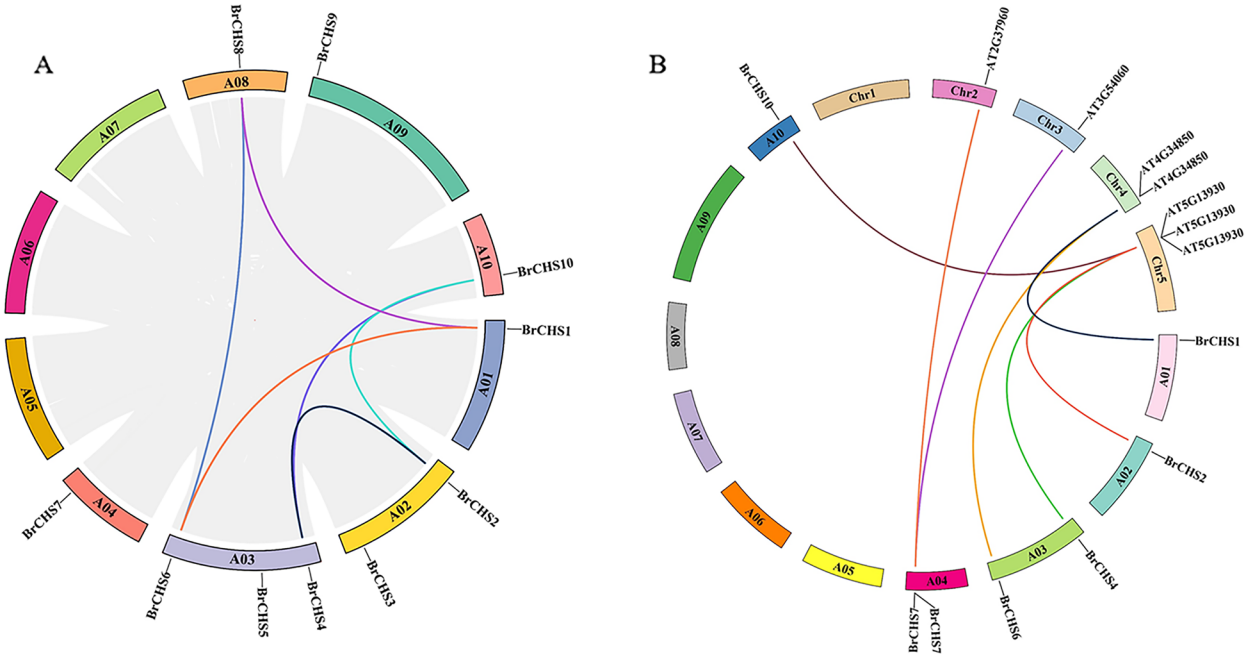


Fig. 5 Collinearity analysis. **A** Collinearity of the Chinese cabbage *CHS* gene family. **B** Collinearity of the Chinese cabbage and Arabidopsis *CHS* gene families

Table 3 Ka/Ks analysis of the duplicated *BrCHS* paralogs

Duplicated Pairs	Ka	Ks	Ka/Ks	Purifying Selection
<i>BrCHS1-BrCHS6</i>	0.026189	0.338989	0.077255	yes
<i>BrCHS1-BrCHS8</i>	0.016385	0.263455	0.062192	yes
<i>BrCHS2-BrCHS4</i>	0.02704	0.448786	0.060252	yes
<i>BrCHS2-BrCHS10</i>	0.019065	0.428219	0.044521	yes
<i>BrCHS6-BrCHS8</i>	0.027322	0.38046	0.071814	yes
<i>BrCHS4-BrCHS10</i>	0.017883	0.472435	0.037853	yes

with *AtCHS* family genes. *AtCHS* (AT5G13930) is related to *BrCHS2*, *BrCHS4*, and *BrCHS10*. Three *AtCHS*-like genes (AT4G34850) correspond to two *BrCHS*-like genes (*BrCHS1* and *BrCHS6*). These genes might have played crucial roles in the evolution of *CHS* genes. These results suggest that the *BrCHS* genes in Chinese cabbage might originate from orthologous *A. thaliana* genes.

Prediction of the *BrCHS* family protein interaction network
The prediction of the Chinese cabbage CHS proteins interaction network is shown in Fig. 6. All proteins were K-means clustered into two categories. Cluster 1 included *BrCHS1*, *BrCHS5*, *BrCHS6*, *BrCHS7*, *BrCHS8* and Bra010535.1-P (Dihydroflavonol 4-reductase-like1, DRL1). Cluster 2 included *BrCHS10*, *BrCHS2*, *BrCHS3*, *BrCHS4*, *BrCHS9*, Bra007142.1-P (Chalcone isomerase, CHI), Bra009101.1-P (Chalcone isomerase like;

CHIL), Bra014831.1-P (Flavanone 3-hydroxylase, F3H) and Bra027457.1-P (Dihydroflavonol 4-reductase, DFR). In addition to the *BrCHS* protein family, several flavonoid metabolism-related enzymes involved in protein interactions. Research has indicated that CHS forms a homemaker for its enzymatic activity [22]. According to the predicted results, there may be interactions between *BrCHS* proteins in different groups.

Cis-elements in *BrCHS* gene promoters
Analysis of the 2000 bp sequence upstream of the *CHS* gene in Chinese cabbage revealed not only many basic *cis*-acting elements but also light responsive elements, phytohormone response elements, stress response elements, and growth- and development-related elements in the *BrCHS* gene family (Fig. 7A). Among the light responsive elements, 80% of the *BrCHS* genes contained G-box elements, and 90% of the *BrCHS* genes contained TCT-motif elements. Phytohormone response elements include abscisic acid response element ABRE, auxin response elements AuxRR-core and TGA-elemnt, gibberellin response related elements P-box, TATC-box and GARE-motif, salicylic acid response element TCA-element, MeJA response elements CGTCA-motif and TGACG-motif. Among the phytohormone response elements, 90% of the *BrCHS* genes contained MeJA response elements, and 80% contained abscisic acid response elements. Among the stress response elements,

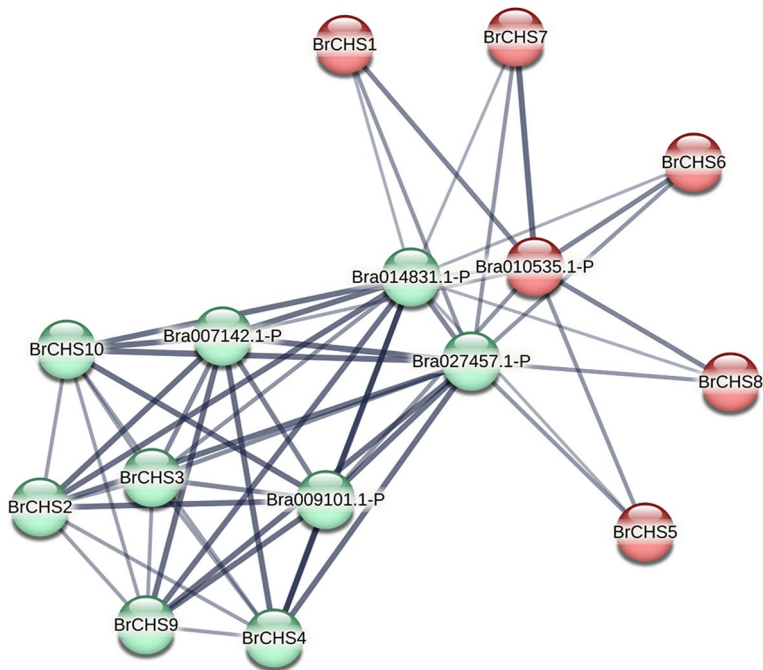
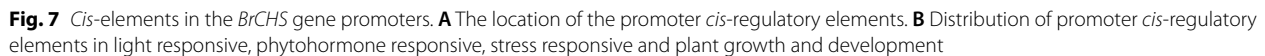


Fig. 6 Interaction network of the *BrCHS* protein family



three abiotic stress treatments. *BrCHS2* and *BrCHS8* responded simultaneously to these three biotic stresses. These findings indicate that most *BrCHS* genes are involved in plant hormone responses and abiotic stress responses.

Functional identification of *BrCHS4* in anthocyanin accumulation

To further determine the biological function of *BrCHS4* in leaf color and anthocyanin accumulation, turnip yellow mosaic virus (TYMV) mediated virus-induced gene silencing (VIGS) technology was applied to silence *BrCHS4* expression in Chinese cabbage. The transient VIGS assay indicated that the pTY-*BrCHS4* Chinese cabbage exhibited green pigmentation, while the bombarded with the empty vector (pTY-S) showed normal pigmentation (Fig. 9A). Moreover, the transcription level of *BrCHS4* was significantly lower in pTY-*BrCHS4* Chinese cabbage than in pTY-S (Fig. 9B). The measurement of flavonoids, anthocyanins and total phenols indicated that *BrCHS4* silencing decreased leaf coloration (Fig. 9C). Furthermore, the results suggested that *BrCHS4* plays an important role in anthocyanin accumulation in Chinese cabbage leaf color.

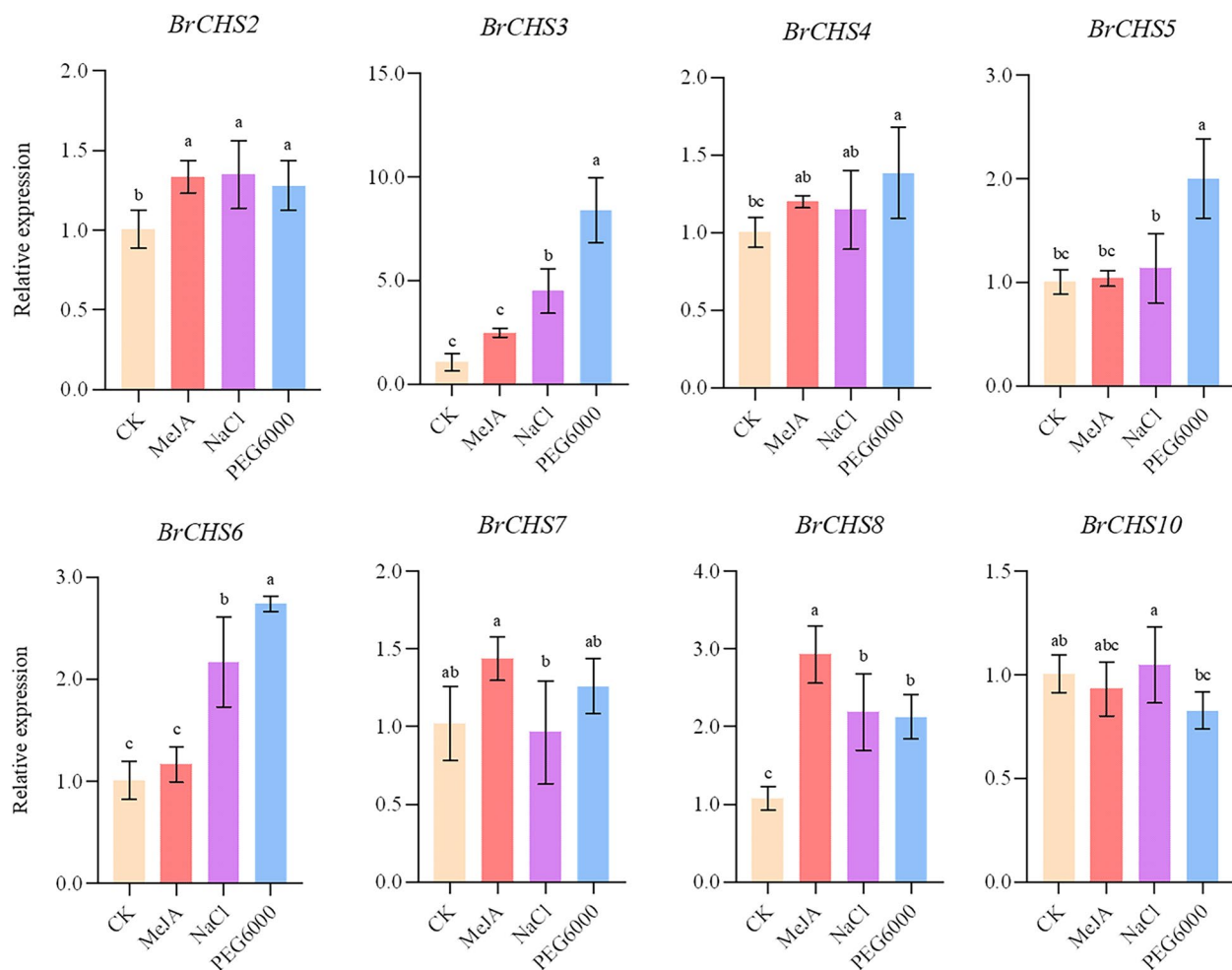


Fig. 8 Expression patterns of *BrCHS* genes in Chinese cabbage under plant hormone treatment and abiotic stress conditions. Chinese cabbage were irrigated with water (CK, control), 0.2 mmol·L⁻¹ methyl jasmonate (MeJA), 400 mmol·L⁻¹ NaCl (salt treatment) and 20% PEG6000 (drought treatment), respectively. All the data indicate means ± SD of three replicates. Different letters represented significant difference in purity ($p < 0.05$)

Discussion

Identification of *BrCHS* family genes

Chalcone synthase (CHS) is a key enzyme in the biosynthetic pathway of phenylpropanoid compounds and is widely present in plants. It plays a role in the initial steps of the phenylpropanoid pathway and condenses three malonyl CoA molecules with one 4-coumaryl CoA molecule to produce naringin chalcone, which is a precursor to various flavonoids [50–53]. After the first plant *CHS* gene was cloned in European celery in 1983 [54], the *CHS* gene was identified in various plants. Most plant genomes contain relatively small *CHS* gene families. For example, 8 *CHS* genes have been identified in tomato [55], 16 members of the *CnCHS* gene family have been identified in *Chrysanthemum nankingense* [56], and 5 *CHS* genes have been identified in mulberry [57]. In this study, 10 *BrCHS* genes were identified from the *CHS* family members of Chinese cabbage at the genome level

using bioinformatics methods. Ten *BrCHS* genes were unevenly distributed on 7 chromosomes of Chinese cabbage, with the largest number of *CHS* genes occurring on chromosome 3. All the *CHS* protein families contained the Chal_Sti_Synt_C and Chal_sti_synt_N conserved domains. The rice [36] *CHS* family is divided into three groups, and 14 maize and 14 soybean *CHS* proteins are divided into 4 groups [11, 13]. According to Koduri et al. [58], the *PpCHS* gene family has 13 members that code for active *CHS* or *CHS*-like enzymes in *Physcomitrella patens*. Phylogenetic analysis revealed that *BrCHS* family members can be divided into two categories (*CHS* and *CHS*-like). Consistent with Guo et al.'s [43] research, *CHS*-like contains the PLN03169 domain. All gymnosperm *CHS*s contain only the PLN03173 domain. Chinese cabbage (Angiosperm) contains structural domains other than PLN03173, including PLN03170, PLN03171, PLN03172, and PLN03173. Previous studies have shown

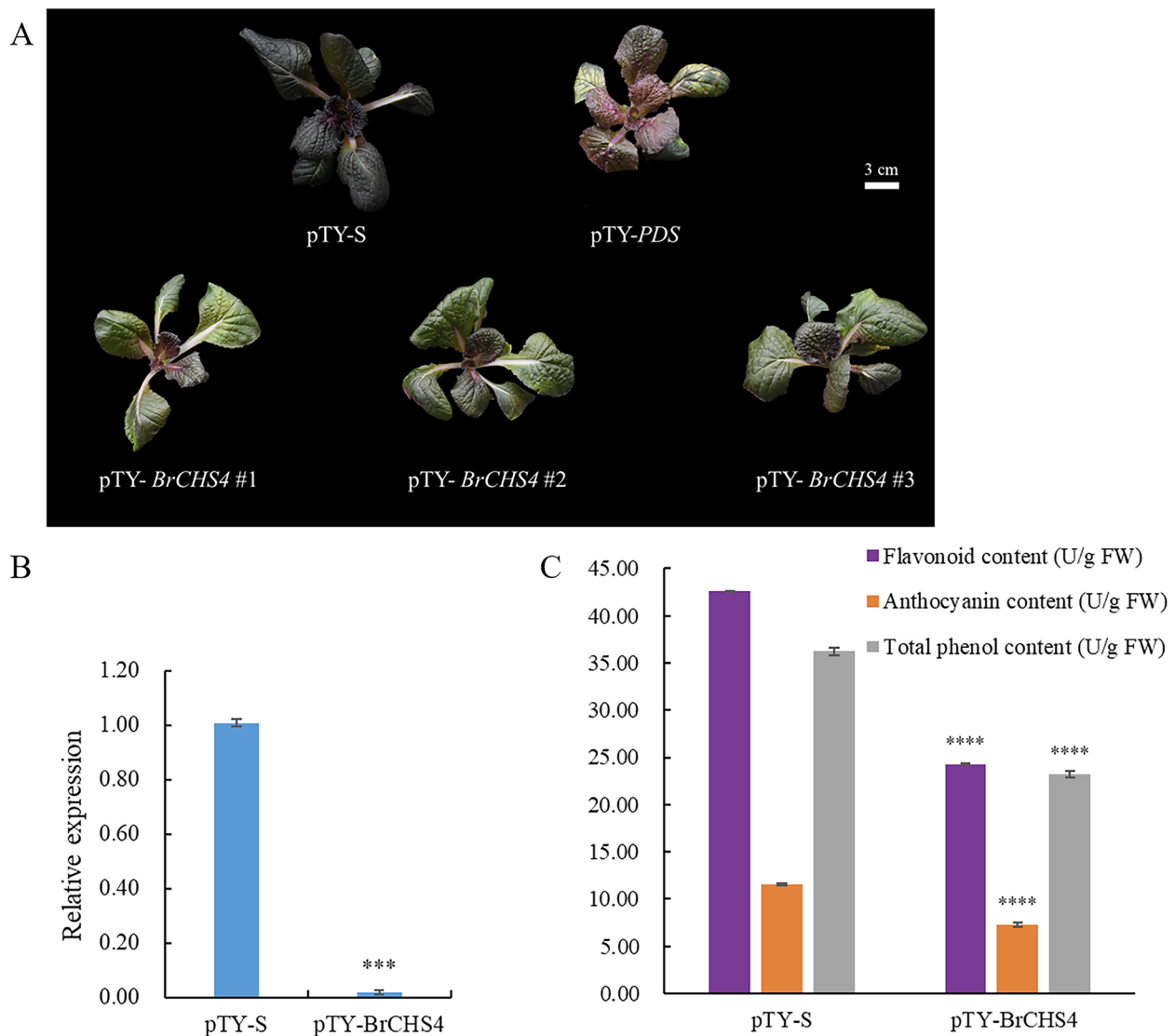


Fig. 9 Virus-induced gene silencing (VIGS) technology to silence *BrCHS4* expression in Chinese cabbage. **A** Phenotypic photos after instantaneous silencing. **B** Transcript analysis of *BrCHS4* in silencing leaves. **C** Measurement of flavonoid, anthocyanin and total phenol content in *BrCHS4* in silencing leaves. All the data indicate means $\pm$ SD of three replicates. Asterisks indicate statistical significance (****, $p < 0.0001$) compared with corresponding controls

that the CHS of the vast majority of plants is localized in the cytoplasm [59], which is consistent with the subcellular localization prediction of 7 out of 10 Chinese cabbage CHS proteins in this study.

Duplication events in CHS family genes

Gene duplication (segment duplication and tandem duplication) is the key force for gene family amplification in the plant genome, and replication events generally occur within groups [60]. This study showed that 10 *BrCHS*s of Chinese cabbage participated in 6 fragment replication events without tandem duplication events.

This finding is consistent with the findings of Han et al. [37] on corn and He et al. [61] on passion fruit (*Passiflora edulis Sims*). In corn, two gene pairs (*ZmCHS01/02* and *ZmCHS13/14*) were found to be involved in segmental repeats. Two gene pairs (*PeCHS3/PeCHS6* and *PeCHS7/PeCHS9*) were found to have fragment duplications in passion fruit. Further analysis revealed that the *BrCHS* fragment replication event occurred within the group, and there was no intergroup replication event. Three genes in the CHS class participated in three fragment replication events, and three genes in the CHS-like class also participated in three fragment replication events.

Among the *BrCHS* genes, the Ka/Ks values for the duplicated *BrCHS* gene pairs ranged from 0.038 to 0.077, indicating that these genes may have acquired limited functional divergence following duplication events [42]. These findings revealed that the *CHS* family is conserved in Chinese cabbage.

***Cis*-acting elements and gene expression patterns of the *CHS* family**

The presence of *cis*-acting elements is indispensable for the proper functioning of various biological and molecular processes in plant cells [62]. The prediction analysis of *cis*-acting elements of the *BrCHS* promoter showed that the *BrCHS* promoter region contained hormone response elements such as MeJA, salicylic acid, auxin, gibberellin, and abscisic acid, which suggested that the expression of *BrCHS* might be affected by stimulation. Wang et al. confirmed that MeJA could increase the expression of *CitCHS* genes (Citrus) [63]. Second, the *BrCHS* promoter region contains some defense stress response elements and elements related to the regulation of plant growth and development. It is predicted that *BrCHS* can improve the ability of Chinese cabbage to resist stress and pathogen infection and participate in the regulation of Chinese cabbage growth and development. The *CHS* promoter region of rice also contains functional elements involved in defense and stress responses, such as flavonoid synthesis, drought induction, and hypoxia induction [64]. Chen et al. revealed that *EaCHS1* (*Eupatorium adenophorum*) enhances plant tolerance to high salt stress in tobacco [65]. *NtCHS1* and *NtCHS3* are induced by both high salt and drought stress in soybean plants [23]. The *CHS* gene family of licorice (*Glycyrrhiza uralensis*) is also involved in plant responses to salt stress and drought [66]. Furthermore, the *BrCHS* promoter region contains *cis* acting elements such as meristem expression and endosperm expression, which are consistent with some promoter elements of the *CHS* gene in buckwheat [67], suggesting that *BrCHS* may be involved in the regulation of Chinese cabbage growth and development. Genes exhibiting similar expression patterns may possess shared characteristics, be regulated by common genetic factors, or trace their origins back to a common ancestor, thereby constituting an indispensable component of genetic research [68, 69]. During their life cycles, plants often face various environmental challenges such as drought, low temperature, and excessive salt, which can significantly impede their growth and development. In response to these adversities, a set of stress-responsive genes are activated to enable plants to adapt effectively [70]. In previous studies, chalcone synthase (*CHS*) is a key enzyme involved in the biosynthesis of flavonoids. Required for the accumulation of purple

anthocyanins in leaves and stems. Our experiments demonstrated that *CHS* gene expression was proportional to the amount of flavonoid substances (Fig. 9). Flavonoids are widely involved in abiotic stress in plants. Studies have shown that high accumulation of flavonoids is a key factor in improving plant resistance. Anthocyanins can directly remove reactive oxygen species in plants, thereby improving plant tolerance to a series of environmental stresses such as salt, drought and heat treatment [65, 71]. Quantitative real-time PCR analysis indicated that four Chinese cabbage *CHS* genes were upregulated under salt stress, while six *BrCHS* genes were upregulated under drought stress. These results indicate that most *BrCHS* genes are induced by drought and salt stress. Although there were MeJA-related elements in 9 *BrCHS* genes, there was no change in gene expression under MeJA hormone treatment except *BrCHS2* and *BrCHS8*. This may be due to variations in the core region of the element and very complex element binding (such as DNA methylation), or MeJA-associated receptors within the cell may be inactivated [9, 72, 73]. To sum up, consistent with the predicted results, the *BrCHS* family may play roles in hormone regulation and stress response.

Function of *CHS* gene in anthocyanin accumulation

CHS has been well studied in the context of anthocyanin synthesis and accumulation, and several reports have described the effects of overexpressing or silencing *CHS* in plants. The first characteristic *chs* mutant was the photowhite mutant of *Antirrhinum majus*. The single *CHS* gene was knocked out and the anthocyanins and flavonoids were lacking [11]. *StCHS* gene overexpression in potato (*Solanum tuberosum*) resulted in anthocyanin accumulation in potato tubers [74]. Many studies in plants have focused on the silencing of *CHS* genes resulting in lower anthocyanin levels in flowers. In this study, we silenced *BrCHS4* gene in Chinese cabbage leaves and found that the leaves of silenced plants changed from purple to green, and the contents of flavonoids, anthocyanins and total phenols were significantly reduced [75–77].

Conclusions

In this study, 10 members of the *CHS* gene family were identified from the genome of Chinese cabbage and were distributed on 7 chromosomes. Seven *BrCHS* proteins were predicted to localize to the cytoplasm, and 3 were predicted to localize to the chloroplast. Sequence analysis revealed that the *BrCHS* family comprises highly conserved hydrophilic proteins. These *BrCHS* gene promoters contain many hormones, stress, plant growth regulation, and light-responsive elements. Expression of the *BrCHS* genes responded to hormone and stress

responses in Chinese cabbage. Transient silencing of *BrCHS4* in Chinese cabbage decreased coloration. These results indicated that *BrCHS4* is involved in flavonoid and anthocyanin accumulation in Chinese cabbage.

Abbreviations

CHS	Chalcone synthase
PKS	Polyketide synthase
CAREs	Cis-Acting regulatory elements
MeJA	Methyl jasmonate
Ka/Ks	The nonsynonymous/synonymous ratio

Supplementary Information

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Supplementary Material 1: Supplemental Table 1. RT-qPCR primers for *BrCHS* genes.

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Authors' contributions

J. L., M. L. and J. Z. designed the research. Z. X., S. X., X. J. and Z. Z., carried out experiments and collected and analyzed the data. Z. X. and M. F. wrote the manuscript. J. L., N. L. and L. Y. revised the manuscript. All authors have read and agreed to the published version of the manuscript.

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Data availability

The datasets supporting the conclusions of this article are included within the article and its additional files. Genomic sequences and gene annotation information of *Brassica rapa* and *Brassica oleracea* were downloaded from <https://brassicadb.cn>. Genomic sequences and gene annotation information of *Arabidopsis* were downloaded from <https://www.arabidopsis.org/index.jsp>.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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